

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

NEURODERM LTD.,
Petitioner,

v.

ABBVIE INC.,
Patent Owner.

PGR2022-00040
Patent 11,091,507 B2

Before TINA E. HULSE, DEVON ZASTROW NEWMAN, and
CYNTHIA M. HARDMAN *Administrative Patent Judges*.

HARDMAN, *Administrative Patent Judge*.

JUDGMENT

Final Written Decision

Granting-in-Part Patent Owner's Non-Contingent Motion to Amend
Denying Petitioner's Motion to Strike
Dismissing Patent Owner's Motion to Exclude
35 U.S.C. § 328(a)

I. INTRODUCTION

This is a Final Written Decision in a post-grant review, entered pursuant to 35 U.S.C. § 328(a) and 37 C.F.R. § 42.73. We have jurisdiction under 35 U.S.C. § 6.

Presently pending is Patent Owner AbbVie Inc.’s non-contingent Motion to Amend, wherein Patent Owner requests cancellation of challenged claims 1–17 of U.S. Patent No. 11,091,507 B2 (“the ’507 patent,” Ex. 1001), and proposes substitute claims 18–30. Paper 27 (“Mot.”). Having reviewed the complete trial record, for the reasons discussed below, we grant Patent Owner’s request to cancel claims 1–17 of the ’507 patent, and deny the request to add proposed substitute claims 18–30. Additionally, for the reasons discussed below, we deny Petitioner’s Motion to Strike (Paper 49) and dismiss as moot Patent Owner’s Motion to Exclude (Paper 66).

A. *Procedural History*

Petitioner NeuroDerm Ltd. filed a Petition requesting post-grant review of all claims (claims 1–17) of the ’507 patent. Paper 2. Patent Owner AbbVie Inc. filed a Preliminary Response. Paper 6. With our authorization, Petitioner filed a Preliminary Reply and Patent Owner filed a Preliminary Sur-reply. Papers 7, 8. In view of the then-available preliminary record, we instituted a post-grant review. Paper 9 (“Inst. Dec.”).

Patent Owner then filed a non-contingent Motion to Amend, requesting cancellation of challenged claims 1–17 of the ’507 patent, and proposing substitute claims 18–30. Mot. 3–4. Petitioner filed an Opposition. Paper 36 (“Opp.”). At Patent Owner’s request, we issued Preliminary Guidance on the Motion to Amend. Paper 40 (“Prelim. Guid.”). Thereafter, Patent Owner filed a Reply in support of the Motion to Amend

(Paper 42, “Reply”), along with new Exhibits 2066–2106. Petitioner filed a Sur-Reply. Paper 68 (“Sur-Reply”).

Petitioner filed an authorized Motion to Strike Evidence (Paper 49), to which Patent Owner filed an Opposition (Paper 58). Patent Owner filed a Motion to Exclude Evidence (Paper 66), to which Petitioner filed an Opposition (Paper 71), and Patent Owner filed a Reply (Paper 73).

On August 29, 2023, we conducted an oral hearing, the transcript of which is of record. Paper 76 (“Tr.”).

B. Real Parties in Interest

Petitioner identifies itself and Mitsubishi Tanabe Pharma Corporation as the real parties in interest. Pet. 108. Patent Owner identifies itself as the real party in interest. Paper 13 (Patent Owner’s Updated Mandatory Notices), 1.

C. Related Matters

Patent Owner identifies pending U.S. Patent Application Nos. 17/378,321 and 17/971,836 as related to the ’507 patent. Paper 13, 1. The parties do not identify any other related matters. *See id.*; Pet. 108.

D. The ’507 Patent (Ex. 1001)

The ’507 patent, titled “Methods of Treating Parkinson’s Disease,” “relates to (a) carbidopa prodrugs, (b) L-dopa prodrugs, (c) pharmaceutical combinations and compositions comprising a carbidopa prodrug and/or an L-dopa prodrug, and (d) methods of treating Parkinson’s disease and associated conditions comprising administering a carbidopa prodrug and an

L-dopa prodrug to a subject with Parkinson's disease.”¹ Ex. 1001, code (54), 1:19–25.

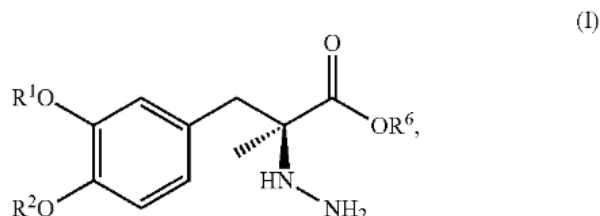
“Parkinson's disease is a chronic and progressive neurodegenerative condition characterized by reduced levels in the brain of the neurotransmitter dopamine.” *Id.* at 1:29–31. “Administration of L-dopa . . . currently is the most effective therapy for treating a patient with Parkinson's disease.” *Id.* at 1:32–34. However, due to premature peripheral metabolism of levodopa throughout the body, a relatively large levodopa dose is required, which can cause complications such as nausea. *Id.* at 1:55–63. “To overcome these problems, L-dopa generally is co-administered with a peripheral aromatic L-amino acid decarboxylase inhibitor such as carbidopa.” *Id.* at 1:58–2:9.

The Specification states that “[e]ven when L-dopa and carbidopa are co-administered, however, it is difficult to consistently maintain the desired dopamine levels in the brain due to the relatively short half-life of L-dopa in plasma.” *Id.* at 2:16–19. “One approach that has been effective in reducing variability of dopamine levels is the continuous intestinal delivery of an adjustable dose of an L-dopa/carbidopa gel known by its commercial name, DuoDopa® in Europe and Duopa® in the United States.” *Id.* at 2:16–35. Achieving continuous intestinal delivery, however, requires a surgical procedure. *Id.* at 2:16–35. The Specification further states that “[a] major challenge to the development of less invasive or otherwise improved modes of administering L-dopa and carbidopa has been the solubility of those

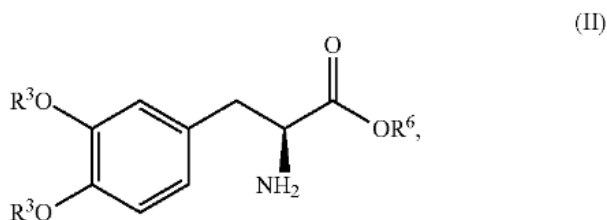
¹ “L-dopa” is shorthand for levodopa. *See* Pet. 10. Prodrugs are “pharmacologically inactive molecules that convert to the active drug in a certain part of the body through metabolic or chemically-mediated processes.” *Id.* at 21; *see also* Ex. 1004 (Sloan Decl.) ¶¶ 26–27; Opp. 5.

compounds.” *Id.* at 2:40–42. The ’507 patent thus states that “[s]table, more highly concentrated, and/or less viscous formulations comprising L-dopa and/or carbidopa (or compounds capable of in vivo bioconversion to L-dopa and/or carbidopa) are desirable.” *Id.* at 2:44–47.

The ’507 patent discloses compositions comprising “L-dopa and carbidopa prodrugs that convert to L-dopa and carbidopa in vivo[,] which allows for delivery by continuous administration methods including intragastric, intramuscular, intravenous, and subcutaneous administration.” *Id.* at 7:8–13. Regarding carbidopa prodrugs, the Specification provides compounds having the following Formula (I):



Id. at 3:3–20. In Formula (I) reproduced above, “R¹ and R² are each independently selected from the group consisting of hydrogen, –P(O)(OH)₂, and –R⁵–O–P(O)(OH)₂; R⁵ is a C₁–C₄-alkyl; R₆ is hydrogen or a C₁–C₄-alkyl; and provided that at least one of R¹ and R² is –P(O)(OH)₂ or –R⁵–O–P(O)(OH)₂.” *Id.* Regarding levodopa prodrugs, the Specification provides compounds having the following Formula (II):



Id. at 3:21–39.² In Formula (II) reproduced above, “R³ and R⁴ are each independently selected from the group consisting of hydrogen, –P(O)(OH)₂, and –R⁵–O–P(O)(OH)₂; R⁵ is a C₁–C₄–alkyl; R⁶ is hydrogen or a C₁–C₄–alkyl; and provided that at least one of R³ and R⁴ is –P(O)(OH)₂ or –R⁵–O–P(O)(OH)₂.” *Id.*

The ’507 patent states that, in one aspect, the disclosure regards a method of treating Parkinson’s disease or an associated condition in a patient comprising administering to the patient a therapeutically effective amount of a pharmaceutical combination comprising a first compound corresponding in structure to Formula (I), or a pharmaceutically acceptable salt thereof, and a second compound corresponding in structure to Formula (II), or a pharmaceutically acceptable salt thereof.

Id. at 3:54–62. In Example 16, “[t]he in vivo conversion of the L-dopa phosphate prodrugs to L-dopa and the carbidopa phosphate prodrugs to carbidopa was evaluated in a rat pharmacokinetics study in which the prodrug was administered intravenously or subcutaneously to the rat.” *Id.* at 82:34–41. In Example 19, “[t]he in vivo conversion of the carbidopa diphosphate to carbidopa was evaluated in a mini-pig pharmacokinetics study in which the prodrug was administered in aqueous solution subcutaneously to a group of three mini-pigs.” *Id.* at 85:54–61. Example 23 provides a study of local irritation and pain at the injection site in rabbits following “intravenous, paravenous, and subcutaneous bolus injection of LD/CD³ diphosphate at concentrations for 200/50 mg/mL.” *Id.* at 88:24–30.

² The structural formula appears to contain a typographical error, in that the lower R³ should instead be R⁴. *See* Ex. 1001, 18:6–60; Pet. 23 n.84.

³ LD and CD are abbreviations for levodopa and carbidopa, respectively. *See, e.g.*, Ex. 1003 (LeWitt Decl.) ¶ 69.

Example 24 provides a “human prediction” of steady state exposures of levodopa, carbidopa, and prodrugs thereof. *See id.* at 89:56–90:51.

E. Substitute Claims

Patent Owner’s non-contingent Motion to Amend proposes to replace challenged claims 1–17 with respective proposed substitute claims 18–30. Mot. 1, 14–15. We reproduce proposed substitute claim 18, the only independent substitute claim, below. The additions to original claim 1 are underlined, and the deletions are shown in double brackets.

18. A method of delivering dopamine to a Parkinson’s disease patient in a pharmacokinetically-consistent manner comprising, continuously subcutaneously administering an aqueous pharmaceutical formulation to the patient, the formulation comprising:

(a) at least about 5 mg/mL of a compound comprising levodopa, wherein the compound comprising levodopa is levodopa 3’-monophosphate, levodopa 4’-monophosphate, levodopa 3’, 4’-diphosphate, levodopa, or a pharmaceutically acceptable salt thereof.

(b) carbidopa, or a pharmaceutically acceptable salt thereof, and

(c) an aqueous carrier,

wherein the method is sufficient to achieve a steady state levodopa plasma level of about 1000 ng/ml to about [[5000]] 3000 ng/ml in the patient for at least 8 hours without co-administration of a catechol-O-methyl transferase inhibitor, and at about 10 hours after the start of said subcutaneous administration the levodopa plasma level is at said steady state.⁴

⁴ In our Preliminary Guidance, we noted that limitation [18a] ends with a period, and “encourage[d] Patent Owner to address this issue if Patent Owner files a Revised Motion to Amend.” Prelim. Guid. 4 n.1. Patent Owner did not file a Revised Motion to Amend. Nevertheless, Patent Owner’s declarant Dr. Jenner suggests that Patent Owner changed this

Id. at App’x A, 1. Substitute claim 19 recites that the steady state levodopa plasma level is achieved without an oral medication for Parkinson’s Disease. *Id.* at 1–2. Substitute claims 20–22 and 26 recite steady state levodopa plasma levels. *Id.* at 2–3. Substitute claims 23–25 and 27–30 recite dosages for “the compound comprising levodopa” or for the carbidopa. *Id.* at 2–4.

II. ANALYSIS

A. *Post-Grant Review Eligibility*

The post-grant review provisions apply only to patents that contain a claim with an effective filing date on or after March 16, 2013. *See* Pub. L. No. 112-29, 125 Stat. 284 (2011), §§ 3(n)(1), 6(f)(2)(A).

The ’507 patent issued from an application filed on May 4, 2020, and claims its earliest priority to a provisional application filed on October 21, 2014. *See* Ex. 1001, codes (22), (60). Because the earliest possible priority date is after March 16, 2013, the ’507 patent is eligible for post-grant review. *See* Inst. Dec. 12. Patent Owner does not dispute this eligibility.

B. *Level of Ordinary Skill in the Art*

In our Institution Decision, we determined that a person of ordinary skill in the art (sometimes abbreviated herein as “POSA”) would have been a member of a team that includes person(s) with

(a) a Ph.D. in a field related to the development of pharmaceuticals (for example, pharmacology, pharmacokinetics, pharmaceuticals, pharmaceutical chemistry, organic chemistry, biochemistry, molecular biology, medicinal chemistry, formulation development, or a related field), or

period to a comma in “Paper 40 at 16.” *See* Ex. 2088 (Jenner Decl.) ¶ 116 n.3. Paper 40 is our Preliminary Guidance, and it does not have a page 16. We are not aware of where in the record, if at all, Patent Owner made this alleged correction.

equivalent degree, and 2–4 years of experience in the analysis of pharmacokinetic data, pharmaceutical formulation development, and/or drug discovery or development, or a B.S. or M.S. degree in a field related to the development of pharmaceuticals, or equivalent degree, and 4–6 years of experience in the analysis of pharmacokinetic data, pharmaceutical formulation development, and/or drug discovery or development, and

(b) a medical degree or equivalent degree with experience in treating patients with Parkinson’s disease.

Inst. Dec. 15–16.

Patent Owner accepts this definition for purposes of its Motion to Amend. Mot. 4. Petitioner does not oppose this definition. *See generally* Opp. We adhere to this level of skill for the reasons set forth in our Institution Decision. *See* Inst. Dec. 12–16.

Based on their statements of qualifications and curricula vitae, for purposes of this Decision we find Petitioner’s declarants Drs. LeWitt and Sloan, and Patent Owner’s declarants Drs. Rosenbaum, Fasano, and Jenner, qualified to provide technical opinions from the perspective of a person of ordinary skill in the art. *See* Ex. 1003 (LeWitt Decl.) ¶¶ 1–10, pp. 87–119 (Dr. LeWitt’s statement of qualifications and curriculum vitae); Ex. 1004 (Sloan Decl.) ¶¶ 1–11, pp. 45–71 (Dr. Sloan’s statement of qualifications and curriculum vitae); Ex. 2086 (Rosenbaum Decl.) ¶¶ 4–9, pp. 1–8 (Dr. Rosenbaum’s statement of qualifications and curriculum vitae); Ex. 2087 (Fasano Decl.) ¶¶ 3–13, App’x A (Dr. Fasano’s statement of qualifications and curriculum vitae); Ex. 2088 (Jenner Decl.) ¶¶ 3–11, App’x A (Dr. Jenner’s statement of qualifications and curriculum vitae).

C. Claim Construction

We interpret a claim “using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. 282(b).” 37 C.F.R. § 42.200(b). Under this standard, we construe the claim “in accordance with the ordinary and customary meaning of such claim as understood by one of ordinary skill in the art and the prosecution history pertaining to the patent.” *Id.*

In our Institution Decision, we provided preliminary constructions for the terms “a compound comprising levodopa” and “patient.” Inst. Dec. 16. Patent Owner contends that, in view of the proposed claim amendments, no construction is needed for “a compound comprising levodopa.” Mot. 4–5. Instead of construing the term “patient,” Patent Owner contends that we should construe the longer term “Parkinson’s disease patient,” and proposes a construction for that term. *Id.* at 5. Petitioner does not expressly address claim construction in connection with the Motion to Amend. *See generally* Opp.

For purposes of deciding the Motion to Amend, we conclude that no claim term requires express interpretation. *See Realtime Data, LLC v. Iancu*, 912 F.3d 1368, 1375 (Fed. Cir. 2019) (“The Board is required to construe ‘only those terms . . . that are in controversy, and only to the extent necessary to resolve the controversy.’”) (quoting *Vivid Techs., Inc. v. Am. Sci. & Eng’g, Inc.*, 200 F.3d 795, 803 (Fed. Cir. 1999)).

D. Legal Principles

In a post-grant review, amended claims are not added to a patent as of right, but rather must be proposed as a part of a motion to amend. 35 U.S.C. § 326(d). A patent owner bears the burden to show by a preponderance of the evidence that the proposed amendment complies with the relevant

regulatory and statutory requirements. *See Sisvel Int’l S.A. v. Sierra Wireless, Inc.*, 81 F.4th 1231, 1238 (Fed. Cir. 2023); 35 U.S.C. § 326(d); 37 C.F.R. § 42.221(d)(1); *Lectrosonics, Inc. v. Zaxcom, Inc.*, IPR2018-01129, Paper 15 at 4–8 (PTAB Feb. 25, 2019) (precedential). Specifically, a patent owner must demonstrate by a preponderance of the evidence that: (1) the amendment proposes a reasonable number of substitute claims; (2) the proposed substitute claims are supported in the original disclosure (and any earlier-filed disclosure for which the benefit of filing date is sought); (3) the amendment responds to a ground of unpatentability involved in the trial; and (4) the amendment does not seek to enlarge the scope of the claims or introduce new subject matter. *See* 35 U.S.C. § 326(d); 37 C.F.R. § 42.221(d)(1).

A petitioner bears the burden to show by a preponderance of the evidence that any proposed substitute claim is unpatentable. *See Sisvel Int’l S.A.*, 81 F.4th at 1238; 37 C.F.R. § 42.221(d)(2); *see also Lectrosonics*, Paper 15 at 3–4.

The written description requirement set forth in 35 U.S.C. § 112(a) ensures that as of the filing date “the inventor actually invented the invention claimed.” *Ariad Pharms., Inc. v. Eli Lilly & Co.*, 598 F.3d 1336, 1350–51 (Fed. Cir. 2010) (en banc). The test “requires an objective inquiry into the four corners of the specification from the perspective of a person of ordinary skill in the art.” *Id.* at 1351. “The written description requirement is satisfied only if the inventor conveys with reasonable clarity to those skilled in the art that, as of the filing date sought, he or she was in possession of the invention, and demonstrates that by disclosure in the specification of the patent.” *Biogen Int’l GmbH v. Mylan Pharms. Inc.*, 18 F.4th 1333, 1341–42 (Fed. Cir. 2021) (cleaned up; citations omitted). “A precise definition of the

invention is pivotal to establishing possession.” *Id.* “An applicant may show possession of the claimed invention by describing it with all of its limitations using ‘such descriptive means as words, structures, figures, diagrams, formulas, etc.’” *Id.* (quoting *Lockwood v. Am. Airlines, Inc.*, 107 F.3d 1565, 1572 (Fed. Cir. 1997)).

“Whether a claimed invention satisfies the written-description requirement of § 112 will depend on the nature of the invention.” *Id.* “[T]he written-description analysis is highly dependent on the facts of each case,” and in general, “is judged based on the state of the art as of the priority date.” *Id.* (quoting *Amgen Inc. v. Sanofi*, 872 F.3d 1367, 1373–74 (Fed. Cir. 2017)).

E. Motion to Amend Statutory and Regulatory Requirements

As noted above (*supra* Section II.D), Patent Owner bears the burden to show by a preponderance of the evidence that the proposed amendment complies with the relevant regulatory and statutory requirements. We address each requirement in turn below.

1. Reasonable Number of Substitute Claims

A motion to amend must “propose a reasonable number of substitute claims.” 35 U.S.C. § 326(d)(1)(B); *see also* 37 C.F.R. § 42.221(a)(3). “There is a rebuttable presumption that a reasonable number of substitute claims per challenged claim is one (1) substitute claim.” *Lectrosonics*, Paper 15 at 4; *see also* 37 C.F.R. § 42.221(a)(3).

Patent Owner proposes thirteen substitute claims to replace the seventeen challenged claims, a less than one-to-one substitution. *See* Mot. 6. Petitioner does not argue that Patent Owner fails to propose a reasonable number of substitute claims. *See generally* Opp.

Based on the above, we determine that Patent Owner meets the requirement of proposing a reasonable number of substitute claims.

2. *Respond to a Ground of Unpatentability*

“A motion to amend may be denied where . . . [t]he amendment does not respond to a ground of unpatentability involved in the trial.” 37 C.F.R. § 42.221(a)(2)(i).

Patent Owner proposes narrowing amendments it contends are responsive to Petitioner’s challenges, including Petitioner’s enablement challenge. *See* Mot. 9–13. Petitioner does not argue that the amendments fail to respond to a ground of unpatentability involved in the trial. *See generally* Opp.

We find that Patent Owner proposes narrowing amendments that are responsive to a ground of unpatentability involved in the trial. For example, in determining in our Institution Decision that Petitioner had shown it was more likely than not that claims 1–17 lack enablement, we cited the breadth of the claim term “a compound comprising levodopa,” which we found included “*any* molecule with a levodopa core moiety, including . . . prodrugs.” Inst. Dec. 30. Patent Owner narrows that term to now recite only four specific compounds and pharmaceutically acceptable salts thereof. *See* Mot. 9.

3. *Scope of the Claims*

A motion to amend may be denied where the amendment seeks to “enlarge the scope of the claims.” 35 U.S.C. § 326(d)(3); 37 C.F.R. § 42.221(a)(2)(ii). Here, proposed substitute independent claim 18 includes narrowing limitations as compared to corresponding original independent claim 1, including with respect to claim term “a compound comprising

levodopa.” *See* Mot. 14. Petitioner does not argue otherwise. *See generally* Opp.

Because the proposed substitute claims include narrowing amendments, we find that Patent Owner does not seek to enlarge the scope of the claims.

4. *New Matter*

A motion to amend may be denied where the amendment seeks to “introduce new matter.” 35 U.S.C. § 326(d)(3); 37 C.F.R. § 42.221(a)(2)(ii). Therefore, a motion to amend must “set forth written description support in the originally filed disclosure of the subject patent for each proposed substitute claim, and also set forth support in an earlier filed disclosure for each claim for which benefit of the filing date of the earlier filed disclosure is sought.” *Lectrosonics*, Paper 15, 7; *see also* 37 C.F.R. § 42.221(b). To address this requirement, Patent Owner provides a claim chart citing various paragraphs of Application No. 14/919,418 (“the ’418 Application”).⁵ *See* Mot. 15–16.

⁵ Petitioner argues that Patent Owner “cites to the ’418 Application, filed October 21, 2015, but not to Application No. 16/866,400 in breach of the requirement to ‘set forth written description support in the *originally filed disclosure* of the subject patent.’” Opp. 3–4, n.1 (quoting *Lectrosonics*, Paper 15, 7). The ’418 Application is related to Application No. 16/866,400 (“the ’400 Application”), which led directly to the ’507 patent, through a series of continuation applications. *See* Ex. 1001, codes (21), (22), (63). On this record, we find Patent Owner’s failure to cite the ’400 Application to be harmless error, given that Petitioner does not argue, and we do not perceive, any material difference between the disclosures of the ’400 Application and the ’418 Application. For simplicity, our references to the “application” herein collectively include the disclosure of both the ’418 Application and ’400 Application. Where we cite only the ’418 Application or only the ’400

For the reasons discussed below, on the full trial record, we find that Patent Owner has not carried its burden of showing by a preponderance of the evidence that the proposed substitute claims introduce no new matter relative to the original disclosure. 35 U.S.C. § 326(d)(3); 37 C.F.R. § 42.221(d)(1); *see also* Paper 58, 2 (Patent Owner acknowledging that it “bears the burden to meet the regulatory and statutory requirements” for a motion to amend, including demonstrating that the substitute claims do not “introduce new matter”).

Importantly, we note that Patent Owner fails to explain how the material cited in the claim chart supports the claims. *See* Mot. 15–16. Instead, Patent Owner provides only a claim chart with bare citations, leaving it to the panel to discern how the cited material supports the claim limitations. *See id.* Patent Owner’s failure to explain the citations constitutes a fatal error, as we have reviewed the passages cited by Patent Owner and are unable to discern where support is found for several of the substitute claim limitations. Where, as here, “the Board is unable to determine how the specification and drawings support the proposed substitute claims, the motion to amend may be denied.” Patent Trial and Appeal Board Consolidated Trial Practice Guide (Nov. 2019) (“TPG”),⁶ 71.

For example, limitation [18a] in proposed substitute claim 18 recites, in relevant part, a formulation comprising “at least about 5 mg/mL of a compound comprising levodopa, wherein the compound comprising

Application, we do so for efficiency, with the understanding that both applications contain the same disclosure. Additionally, we use the terms “application,” “specification,” and “disclosure” interchangeably herein.

⁶ Available at <https://www.uspto.gov/TrialPracticeGuideConsolidated>.

levodopa” is levodopa itself or one of three recited levodopa prodrugs. Mot. App’x A, 1 (limitation [18a]). Although the proposed amendment to this claim narrows the genus of “compound[s] comprising levodopa” to four specific compounds, like original claim 1 of the ’507 patent, this limitation still embraces levodopa itself. We previously observed that the application’s “singular focus on L-dopa prodrugs fails to support the claims’ option of using L-dopa itself.” Inst. Dec. 23 (quoting Pet. 66).

In its Motion to Amend, as support for claim limitation [18a], Patent Owner provides a list of citations to the ’418 Application⁷ (namely, paragraphs 5, 8, 45, 74, 76–78, 165, 300–13, 329–40, 391–417). Mot. 15–16. Patent Owner, however, does not explain how this material overcomes the issue we previously identified, i.e., that the application fails to support the claim’s option of using levodopa itself in the claimed methods. Having reviewed each of the aforementioned paragraphs, we again find that they fail to support the claims’ option of using levodopa itself. *See also* Opp. 7–9 (addressing why Patent Owner’s citations do not provide adequate written description support for a formulation comprising levodopa).

In particular, paragraphs 8, 74, 76–78, 165, 300–13, and 329–40 are limited to disclosures directed to prodrugs. *See* Ex. 2060 (’418 Application), 17–18 (¶ 8), 29–30 (¶¶ 74, 76–78), 53–54 (¶ 165), 81–86 (¶¶ 300–13), 93–100 (¶¶ 329–40). Paragraph 45 defines “pharmaceutically acceptable salt,” with no mention of carbidopa or levodopa. *See id.* at 22 (¶ 45). Paragraphs 391–417 encompass Examples 16–24, none of which relate to administration of a formulation comprising levodopa together with carbidopa, as required

⁷ A copy of the specification of the ’418 Application as filed is located in the record at Exhibit 2060, pages 15–168.

by limitations [18a] and [18b]. *See id.* at 121–31 (¶¶ 391–417); *see also* Opp. 7–8 (explaining why Examples 16 and 19 do not support the proposed amended claims), 9 (explaining that the remaining examples “only disclose administration (or predictions) of formulations comprising **prodrugs** of both levodopa and carbidopa, not the claimed formulations”).

Paragraph 5 of the ’418 Application at least mentions levodopa, stating (in the Background of the Invention Section), “Stable, more highly concentrated, and/or less viscous formulations comprising L-dopa and/or carbidopa (or compounds capable of *in vivo* bioconversion to L-dopa and/or carbidopa) are desirable.” Ex. 2060 (’418 Application), 16 (¶ 5). Although this sentence mentions levodopa, we previously observed that, in context, the sentence does not appear to be characterizing or identifying the invention. *See* Inst. Dec. 25. Instead, it merely states what is “desirable.” Beyond this sentence in paragraph 5, the remainder of the application, including the “Field of the Invention” and “Summary of the Invention” sections, discusses levodopa and carbidopa prodrugs, not levodopa and carbidopa themselves.⁸ *See id.*; Pet. 22–27 (discussing how the patent’s disclosure focuses on levodopa and carbidopa prodrugs).

We also find that Patent Owner’s Motion to Amend fails to identify adequate written description support for claim limitation [18b], which recites that the claimed formulation comprises carbidopa. In our Institution Decision, we observed that the application “generally focuses on the use of carbidopa prodrugs, not carbidopa itself, in the disclosed methods and

⁸ To the extent the sentence in paragraph 5 *is* characterizing the invention, the sentence also refers to “compounds capable of *in vivo* bioconversion to L-dopa and/or carbidopa”—i.e., levodopa and carbidopa prodrugs—which is consistent with the application’s focus on prodrugs.

compositions.” Inst. Dec. 24. In its Motion to Amend, Patent Owner provides a list of citations to the ’418 Application (namely, paragraphs 3, 5, 45, 94, 111, 391 (Table 16-B), 399 (Table 19-A)) as support for claim limitation [18b], but again fails to explain how this material overcomes the problem we originally identified, i.e., that the application fails to support the claim’s option of using carbidopa itself in the claimed formulations. *See* Mot. 16. Having reviewed Patent Owner’s list of citations, we again find that they fail to support the claims’ option of using carbidopa itself. *See also* Opp. 7–9 (addressing why Patent Owner’s citations do not provide adequate written description support for a formulation comprising carbidopa).

Paragraph 3 discusses the state of the art, noting that levodopa is generally co-administered with carbidopa. *See* Ex. 2060 (’418 Application), 15–16 (¶ 3). Patent Owner does not explain how this information indicates that Patent Owner’s invention extends to a formulation comprising these drugs. As to paragraph 5, we reiterate our observation in our Institution Decision (and stated above) that although this paragraph mentions carbidopa, in context it does not appear to be characterizing or identifying the invention. Inst. Dec. 25; Ex. 2060 (’418 Application), 16 (¶ 5). Paragraph 45 defines “pharmaceutically acceptable salt,” with no mention of carbidopa or levodopa. Ex. 2060 (’418 Application), 22 (¶ 45). Paragraph 94 states, in its entirety: “The present disclosure also relates to pharmaceutical combinations and compositions comprising a carbidopa prodrug and/or an L-dopa prodrug.” *Id.* at 35 (¶ 94). There is no suggestion here of co-formulating carbidopa itself with levodopa or a levodopa

prodrug.⁹ Paragraph 111 teaches that “[t]he composition may independently comprise the *first compound* and the *second compound* as either the free form of the compound or a pharmaceutically acceptable salt of the compound.” *Id.* at 40 (¶ 111) (emphasis added). The application defines the terms “first compound” and “second compound” in this sentence to mean carbidopa prodrugs and levodopa prodrugs, respectively. *See id.* at 39–40 (¶ 110); Opp. 9. Patent Owner does not explain how any of the above disclosures—none of which mention carbidopa itself—support a claim to a method of using a formulation comprising carbidopa itself, let alone in combination with levodopa.

Finally, Patent Owner cites portions of Examples 16 and 19 (paragraphs 391 and 399), but in our Institution Decision, we disagreed that these passages support a claim to a method of using a formulation comprising carbidopa itself. Inst. Dec. 25; *see also* Opp. 7–8. The bare citations in the claim chart fail to elucidate how these passages support the claimed subject matter.

⁹ To the extent Patent Owner suggests that the application’s use of “and/or” in the phrase “carbidopa prodrug and/or an L-dopa prodrug” demonstrates possession of a composition comprising a levodopa prodrug with carbidopa, we disagree. First, Patent Owner does not articulate such an argument in its Motion and therefore has forfeited it. *See In re Google Tech. Holdings LLC*, 980 F.3d 858, 862–62 (Fed. Cir. 2020) (indicating that failure to make an argument before the Office compels a finding of forfeiture). Second, even if Patent Owner had made such an argument, we would have rejected it for the reasons articulated in our Institution Decision (*see* Inst. Dec. 26), and because “a description that merely renders the invention obvious does not satisfy the [written description] requirement.” *Ariad Pharms.*, 598 F.3d at 1352.

We also find that Patent Owner’s Motion to Amend fails to identify adequate written description support for limitation [18d], which recites that the claimed method “is sufficient to achieve a steady state levodopa plasma level of about 1000 ng/ml to about 3000 ng/ml in the patient for at least 8 hours.” Mot., App’x A, 1 (limitation [18d]). Patent Owner cites paragraphs 266–68 and 415–17 as support for this limitation. Mot. 16. Paragraphs 266–68 recite that “an amount of the first compound and an amount of the second compound are administered such that in combination they are sufficient to achieve” certain levodopa plasma levels in the patient. *See* Ex. 2060 (’418 Application), 69 (¶¶ 266–68). The application again defines the terms “first compound” and “second compound” in this sentence to mean carbidopa prodrugs and levodopa prodrugs. *See id.* at 68 (¶¶ 264–65); *see also* Opp. 10–11. Patent Owner does not explain how these disclosures—which do not mention levodopa or carbidopa themselves—support a claim to a method of using a formulation comprising these drugs to achieve the claimed pharmacokinetic parameters.

Additionally, Patent Owner fails to explain how paragraphs 266–68 provide support for the claimed steady state levodopa plasma range of “about 1000 ng/ml to about 3000 ng/ml.” Paragraph 266 discloses open-ended ranges, including “at least about 1,000 ng/mL” and “at least about 3,000 ng/mL.” *See* Ex. 2060 (’418 Application), 69 (¶ 266). Paragraph 267 cites closed ranges, but those ranges relate to levodopa plasma levels resulting from administration of a formulation comprising levodopa *prodrugs* and carbidopa *prodrugs*. Additionally, paragraph 267 discloses various ranges that encompass the claimed range, e.g., “about 10 ng/mL to about 9,000 ng/mL” and “about 50 ng/mL to about 4,000 ng/mL,” but does not disclose the claimed range itself. *See id.* at 69–70 (¶ 267). Finally,

paragraph 268 cites time ranges for how long the levodopa concentrations can be maintained, but also does not disclose the claimed levodopa plasma range. *See id.* at 70 (¶ 268).

Given that the claimed range is not disclosed in the application, Patent Owner had to do more than present us with a list of paragraphs in the application in order to explain how a person of ordinary skill in the art would have understood that the application discloses the claimed steady state levodopa plasma range. *See, e.g., Indivior UK Ltd. v. Dr. Reddy's Labs. S.A.*, 18 F.4th 1323, 1328 (Fed. Cir. 2021) (“In the case of a claimed range, a skilled artisan must be able to reasonably discern a disclosure of that range.”). Although the application need not recite the language of the proposed substitute claims verbatim (or “*in haec verba*”) (*see Ariad Pharms.*, 598 F.3d at 1352), under the circumstances here, a mere citation to the application without any explanation as to why a person of ordinary skill in the art would have recognized that the inventors possessed the claimed steady state levodopa plasma range is inadequate. As noted above, “[i]f the Board is unable to determine how the specification and drawings support the proposed substitute claims, the motion to amend may be denied.” TPG 71.

For limitation 18[d], Patent Owner also cites paragraphs 415–17 as providing written description support. Mot. 16. These paragraphs encompass Example 24, which is titled “Human Prediction of Steady State Exposures of L-dopa and L-dopa 4'-monophosphate, Carbidopa and Carbidopa 4'-monophosphate as well as Phosphorus daily load.” Mot. 16; Ex. 2060 ('418 Application), 130 (¶¶ 415–17). Although the bare citations in the claim chart again fail to elucidate how these passages support limitation 18[d], in connection with its enablement arguments, Patent Owner asserts that Example 24 allows persons of ordinary skill in the art “to predict

the steady state levodopa plasma levels for various combinations” of levodopa, carbidopa, and their prodrugs. Mot. 12–13. Even if this statement is correct, on this record, Patent Owner fails to explain how a disclosure that permits a person of ordinary skill in the art to predict the claimed range demonstrates actual possession of that range. “[T]he hallmark of written description is disclosure.” *Ariad Pharms.*, 598 F.3d at 1351. “A written description sufficient to satisfy the requirement of the law requires a statement of an invention, not an invitation to go on a hunting expedition to patch together after the fact a synthetic definition of an invention.” *Indivior UK Ltd.*, 18 F.4th at 1329.

At the oral hearing, when asked where the application supports the range recited in limitation [18d], Patent Owner’s counsel identified Figures 4 and 10 and Examples 17 and 19 of the application. Tr. 35:11–18. Even if counsel is correct that these portions of the application provide written description support for limitation [18d], in its Motion, Patent Owner did not cite these portions of the application. *See* Mot. 16. “During an oral hearing, a party . . . may only present arguments relied upon in the papers previously submitted.” TPG 85–86; *see also* 37 C.F.R. § 42.22 (motion must contain “[a] full statement of the reasons for the relief requested”). Accordingly, we do not consider these portions of the application in ascertaining whether Patent Owner carried its burden of demonstrating that the proposed substitute claims contain no new matter.

Although Patent Owner does not explain how the material cited in the claim chart supports the proposed substitute claims, as we observed in our Preliminary Guidance, elsewhere in the Motion Patent Owner argues that the claims have written description support in the ’507 patent Specification. Prelim. Guid. 5 (citing Mot. 6–9). For example, Patent Owner argues that

Table 16-A in the Specification discloses that levodopa itself was subcutaneously administered to rats, providing levodopa exposure comparable to the exposure of the levodopa 3'-monophosphate prodrug. Mot. 7. Patent Owner also argues that the Specification, including Examples 16 and 19, together teach that levodopa and carbidopa “themselves can be delivered subcutaneously to the same effect” as the prodrugs, while Example 24 teaches that “the animal study data can be used to predict human blood plasma concentrations” for combinations of drugs including levodopa and carbidopa. *Id.* at 8.

We agree with Petitioner that this is not sufficient to carry Patent Owner’s burden. “[D]espite all its citations, [Patent Owner] does not claim the application actually includes an example of the claimed formulations. Instead, [Patent Owner] suggests a skilled artisan could somehow cobble one together based on Examples 16 and 19.” Opp. 9. Additionally, Patent Owner does not contend that the application discloses the claimed plasma concentration range, but rather, asserts that a person of ordinary skill in the art could do calculations to predict the claimed range. Mot. 12–13. We agree with Petitioner that “even if [Patent Owner’s] premise[s] were true, ‘a description that merely renders the invention obvious does not satisfy the [written description] requirement.’” Opp. 10 (quoting *Ariad Pharms.*, 598 F.3d at 1352).

Moreover, despite the apparent disconnects between the limitations of the proposed substitute claims and the disclosure in the application as discussed above, Patent Owner’s Motion to Amend is “based entirely on attorney argument, with no expert opinion in support of its positions.” Opp. 2; *see also id.* at 3–6 (discussing disconnect between the application and Patent Owner’s attorney arguments). Patent Owner correctly contends

that it is not required to submit expert declarations with its Motion to Amend in order to carry its burden. *See* Tr. 42:4–19 (Patent Owner’s counsel arguing that “there’s no specific requirement that we need to put an expert declarations” to show no new matter); TPG 69 (“Although not required, patent owners may offer arguments regarding patentability of any proposed substitute claim and may support their arguments regarding patentability with testimony from a technical expert.”).

Nevertheless, Patent Owner does bear the burden of proving that its proposed amendment satisfies the statutory and regulatory requirements associated with a motion to amend. *See* 35 U.S.C. § 326(d)(3); 37 C.F.R. § 42.221(d)(1); *see also* Paper 58, 2 (Patent Owner acknowledging that it “bears the burden to meet the regulatory and statutory requirements” for a motion to amend). Patent Owner offers only a bare list of citations without any explanation as to how the listed paragraphs disclose the claimed subject matter.¹⁰ *See* Mot. 15–16. Some of the citations are the same as those Patent Owner relied on prior to institution, and we already expressed skepticism in our Institution Decision that those citations supported subject matter that was embraced by the original claims and continues to be embraced by the proposed substitute claims (e.g., a formulation for continuous subcutaneous administration comprising levodopa and carbidopa).

¹⁰ To the extent Patent Owner seeks to rely on the declarations submitted together with its Reply in support of its Motion to Amend to carry its burden of meeting the regulatory and statutory requirements for a motion to amend, we decline to consider those materials. “While replies and sur-replies can help crystalize issues for decision, a reply or sur-reply that raises a new issue or belatedly presents evidence may not be considered.” TPG 74.

Under the circumstances, Patent Owner’s list of citations does not, on its face, adequately demonstrate support for the proposed amended claims. In such case, a denial of the motion to amend for failure to satisfy the statutory and regulatory requirements is appropriate. *See, e.g., B.E. Tech., LLC v. Google, Inc.*, Nos. 2015–1827 through -1829, 2015–1879, 2016 WL 6803057, at *7 (Fed. Cir. 2016) (finding that Board did not err in denying motion to amend where patent owner failed to identify written description support and instead “simply used a string citation to support its proposed substitute limitations”); *see also Fleming v. Cirrus Design Corp.*, 28 F.4th 1214, 1225–26 (Fed. Cir. 2022) (affirming denial of motion to amend where the proposed amended claims were not supported by the portions of the application patent owner cited); *Sisvel Int’l S.A.*, 81 F.4th at 1241–42 (affirming denial of motion to amend where patent owner failed to demonstrate that proposed amended claims did not enlarge scope); TPG 71 (“If the Board is unable to determine how the specification and drawings support the proposed substitute claims, the motion to amend may be denied.”).

For the above reasons, we determine that Patent Owner has not demonstrated by a preponderance of the evidence that the proposed substitute claims introduce no new matter as required by 37 C.F.R. § 42.221(b). Accordingly, we deny Patent Owner’s Motion to Amend to the extent it seeks to add proposed substitute claims 18–30.

F. Petitioner’s Argument on Lack of Adequate Written Description

As an alternative basis to deny Patent Owner’s request to add proposed substitute claims 18–30, we find that Petitioner has demonstrated by a preponderance of the evidence that these claims lack adequate written description support. We address Petitioner’s arguments below.

1. Whether the '418 Application Describes the Claimed Formulation Limitations ([18a-c])

Petitioner argues that the '418 Application contains no disclosure of “**any** aqueous formulation comprising **both** a) carbidopa itself **and** b) levodopa itself or one of levodopa’s three claimed prodrugs,” as limitations [18a–c] of Patent Owner’s proposed amended claim 18 require. Opp. 6; *see also* Pet. 48–56; Ex. 1003 (LeWitt Decl.) ¶ 104; Ex. 1004 (Sloan Decl.) ¶ 79. Petitioner explains how “each section of the specification focuses only on the group of closely-related phosphate prodrugs of Formula (I) (carbidopa prodrugs) and Formula (II) (L-dopa prodrugs),” combinations of those prodrugs, and methods of continuous subcutaneous administration of those combinations of prodrugs. Pet. 49; *see also id.* at 21–27; Ex. 1004 (Sloan Decl.) ¶¶ 38–55, 64–69; Ex. 1003 (LeWitt Decl.) ¶¶ 89–97; Opp. 6–10; Ex. 1047 (Forrest Decl.) ¶¶ 40–74, 80–92.

Patent Owner disputes Petitioner’s written description arguments, arguing that “the specification as a whole, including its detailed discussion of PK [pharmacokinetic] studies, in conjunction with the well-developed prior art on LD and CD, shows possession of not only subcutaneous administration of the four claimed LD phosphate prodrugs or LD with CD, but also of the claimed steady state LD plasma levels.” Reply 2. As to disclosure of the formulation limitations in particular, Patent Owner argues that Petitioner’s “demand for a working example using an LD/CD formulation is legally incorrect.” *Id.* at 4. Patent Owner argues that neither an actual reduction to practice, nor a disclosure repeating the claim limitations verbatim is required. *Id.* (citing *Falko-Gunter Falkner v. Inglis*, 448 F.3d 1357, 1366 (Fed. Cir. 2006); *Ariad Pharms.*, 598 F.3d at 1352).

Patent Owner argues that the examples in the application “test both prodrug and parent compounds” and demonstrate that the prodrugs “rapidly and completely convert to deliver parent LD and CD to the body,” while the “human dose prediction taught in [Example 24 of] the patent validates its teaching that LD and CD administered subcutaneously in solutions of at least about 5 mg/mL of LD can reach desired plasma levels.” Reply 5–6 (citing, *e.g.*, Ex. 2088 (Jenner Decl.)¹¹ ¶¶ 23, 89, 92, 103–05, 110, 122–62; Ex. 2086 (Rosenbaum Decl.) ¶¶ 12–15, 63, 85, 107–10, 114; Ex. 1002 (’400 application¹²), 507–08 (¶¶ 415–17), 498–99 (¶¶ 391–92), 502–03 (¶¶ 399–401)). Patent Owner further argues that “the prior art teaches multiple ways to obtain LD compositions of at least 5 mg/mL, including ‘off the shelf’ solutions, buffered solutions, and dissolved LD salts,” and “[n]o solubilization beyond these methods is required.” Reply 6 (citing Ex. 2088 (Jenner Decl.) ¶¶ 57–81, 169–174; Ex. 2090 (Remenar¹³), 26–28; Ex. 2093 (Nutt¹⁴), 483–84; Ex. 1012 (Quinn¹⁵), 1132; Ex. 2095 (Larson¹⁶) ¶¶ 48–64).

¹¹ Together with its Reply, Patent Owner for the first time in this proceeding submitted technical declarations in support of its arguments. *See generally* Reply; Exhibits 2086–2088. As discussed below (*infra* Section III), Petitioner filed a Motion to Strike these declarations (and other evidence filed with the Reply).

¹² A copy of the specification of the ’400 Application as filed is found within Exhibit 1002, at pages 392–533.

¹³ Remenar et al., WO 2005/023185 A2, published March 17, 2005.

¹⁴ Nutt et al., *The “On-Off” Phenomenon in Parkinson’s Disease*, 310(8) New England Journal of Medicine 483–88 (1984).

¹⁵ Quinn et al., *Control of on/off phenomenon by continuous intravenous infusion of levodopa*, 34 Neurology (Cleveland) 1131–36 (1984).

¹⁶ Larson et al., US 2007/0027216 A1, published February 1, 2007.

After considering all of the arguments and evidence of record, Petitioner persuades us that the '418 Application lacks adequate written description support for the formulation limitations of proposed substitute claim 18 [limitations 18a–8c]. As outlined by Petitioner and its declarants, the '418 Application, including the Abstract, Field of the Invention, Summary of the Invention, and Detailed Description of the Invention sections, uniformly focuses on phosphate prodrugs of levodopa and carbidopa. *See, e.g.*, Ex. 1004 (Sloan Decl.) ¶¶ 38–55, 64–69; Ex. 1003 (LeWitt Decl.) ¶¶ 89–97; Ex. 1047 (Forrest Decl.) ¶¶ 40–74, 80–92; Opp. 6–10; Pet. 23–26, 49–55.

We agree with Petitioner that “nothing in the specification suggests that the invention includes methods of subcutaneous infusion with anything other than the carbidopa prodrugs and L-dopa prodrugs of Formulas (I) and (II), respectively, or teaches how to get sufficient amounts of either L-dopa or carbidopa into solution, or achieve particular L-dopa plasma levels, without those particular prodrugs.” Pet. 27. Petitioner persuades us that the '418 Application fails to contain “[so] much as a *single generic sentence*, anywhere in the application, suggesting the inventors possessed the claimed formulations.” Sur-reply 3. Additionally, the '418 Application contains no example teaching administration of an aqueous formulation comprising carbidopa itself with levodopa itself or one of the three claimed levodopa prodrugs, let alone one that is suitable for continuous subcutaneous administration. *See* Ex. 1047 (Forrest Decl.) ¶¶ 61–72, 82–92.

Patent Owner is correct that to establish written description support, neither “an actual reduction to practice,” nor a verbatim recitation of the claims is required. Sur-reply 4. Nevertheless, “[a]lthough examples are not always required to satisfy the written description requirement, the lack of

any disclosure of examples may be considered when determining whether the claimed invention is adequately described.” *Bos. Sci. Corp. v. Johnson & Johnson*, 647 F.3d 1353, 1364 (Fed. Cir. 2011). Here, the lack of an actual reduction to practice or verbatim description of a formulation that falls within the scope of the claims is relevant, because we also see no other description or material in the ’418 Application that suggests the claimed methods extend to continuous subcutaneous administration of a composition comprising levodopa or one of the three claimed levodopa prodrugs, together with carbidopa, in an aqueous carrier.

Patent Owner argues that the examples provide useful information regarding the ability of levodopa and carbidopa to be administered subcutaneously. For example, Patent Owner argues that Example 16 and Table 16-A disclose that subcutaneous administration of levodopa “provided exposures of levodopa comparable to the exposure of the 3’-monophosphate prodrug of levodopa.” Mot. 6–7 (citing Ex. 1001, 82:34–83:34; Ex. 2061 (Sloan Depo. Tr.), 85:4–11, 89:17–90:3; Ex. 2062 (LeWitt Depo. Tr.), 223:23–224:10); *see also* Reply 5; Ex. 2088 (Jenner Decl.) ¶¶ 103–05, 125, 128–29, 137. Patent Owner further argues that Examples 16 and 19 report *in vivo* studies where “[c]arbidopa ‘[was] administered . . . subcutaneous[ly] to rats.’” Mot. 7 (citing Ex. 2061 (Sloan Depo. Tr.), 85:4–11; Ex. 1001, 82:34–83:43, 85:54–86:46; Ex. 2062 (LeWitt Depo. Tr.), 229:25–230:4, 231:11–14); *see also* Reply 5–6; Ex. 2088 (Jenner Decl.) ¶ 137 (“Table 16-B teaches the subcutaneous administration of unmodified carbidopa, which satisfies the element.”). Patent Owner argues that these examples teach a person of ordinary skill in the art that “the parent compounds themselves can be delivered subcutaneously to the same effect” as the prodrugs. Mot. 8; *see also* Reply 5. Patent Owner also argues that the “prodrug disclosures” in the

application “are relevant across the method,” because the prodrugs “have the same core structures as LD and CD and, as the patent teaches, rapidly and completely convert to deliver parent LD and CD to the body.” Reply 5–6; *see also* Ex. 2088 (Jenner Decl.) ¶¶ 104–05, 125.

We agree with Petitioner that “Examples 16 and 19 indisputably do not disclose a *single* formulation including *both* carbidopa *and* levodopa (or one of three levodopa prodrugs), as required by the substitute claims.” Opp. 7. Instead, the Examples disclose separately administering carbidopa and levodopa as comparators for evaluating the carbidopa and levodopa prodrugs. *Id.* at 7; Ex. 2060 (’418 Application), 121 (¶ 391) (“For comparison, a rat pharmacokinetics study with L-dopa and carbidopa was conducted to help assess the *in vivo* conversion of the prodrugs.”), 125 (¶ 399) (“For comparison, a pharmacokinetics study with carbidopa also was conducted to help assess the *in vivo* conversion of the carbidopa prodrugs.”); Ex. 1047 (Forrest Decl.) ¶¶ 61–62, 65–66, 85; Ex. 1003 (LeWitt Decl.) ¶ 58. We agree with Petitioner that “as to LD and CD specifically, the application discloses their separate use, as controls, in Examples 16 and 19-A, and then never again suggests their administration.” Sur-reply 4.

On this record, we find that the separate administration of levodopa and carbidopa in Examples 16 and 19 is not sufficient to indicate to a person of skill that formulations comprising levodopa and carbidopa themselves are actually part of the invention.¹⁷ Patent Owner’s declarant Dr. Jenner

¹⁷ Although the claims cover the use of carbidopa with levodopa or one of three recited levodopa prodrugs, Petitioner largely focuses on lack of written description support for a formulation comprising carbidopa and levodopa. Because this issue is dispositive, we do the same.

characterizes Examples 16 and 19 as teaching that “continuous subcutaneous administration of levodopa was possible, beneficial, and compatible with carbidopa.” Ex. 2088 (Jenner Decl.) ¶ 135. But the ’418 Application nowhere discloses this conclusion. Even if a person of ordinary skill in the art could have formed this conclusion based on a reading of the ’418 Application as a whole, “a description that merely renders the invention obvious does not satisfy the [written description] requirement.” *Ariad Pharms.*, 598 F.3d at 1352; Opp. 10.

Additionally, we find that use of levodopa and carbidopa in separate examples does not help Patent Owner establish written description support because “[a] patent owner cannot show written description support by picking and choosing claim elements from different embodiments that are never linked together in the specification.” *See Flash-Control, LLC v. Intel Corp.*, No. 2020-2141, 2021 WL 2944592, at *4 (Fed. Cir. July 14, 2021). “The written description requirement is not met when, as here, the specification provides at best disparate disclosures that an artisan might have been able to combine in order to make the claimed invention.” *Id.*

The lack of disclosure of a levodopa/carbidopa formulation for use in the claimed methods is perhaps unsurprising, given the ’418 Application’s express teaching that “[a] major challenge to the development of less invasive or otherwise improved modes of administering L-dopa and carbidopa has been the solubility of those compounds. They each have low aqueous solubility at the pH range required for infusion.” Ex. 2060 (’418 Application), 16 (¶ 5); *see also id.* at 23 (¶ 49) (“[T]he inherently low aqueous solubility of L-dopa and carbidopa at physiologically acceptable pH for infusion presents a significant technical challenge to the development of

improved pharmaceutical compositions and methods of treatment.”);
Pet. 19; Ex. 1003 (LeWitt Decl.) ¶ 48; Ex. 1047 (Forrest Decl.) ¶ 27.

This express teaching is consistent with the prevailing view in the art at the time of the invention that the low aqueous solubility of levodopa and carbidopa at the required pH made it impossible to solubilize sufficient amounts of the drugs to achieve effective continuous subcutaneous administration. *See, e.g.*, Ex. 1013 (Pfeiffer¹⁸), 473 (“For drugs that are very soluble and require small infusion volumes, continuous subcutaneous infusion is a viable treatment option. While this is not possible for L-dopa, it has proven to be feasible for two dopamine agonists, apomorphine and lisuride.”); Ex. 1014 (Lundqvist¹⁹), 340 (“Especially the low aqueous solubility of levodopa made it impossible to consider subcutaneous administration since considerable volumes would have been required.”); Ex. 1003 (LeWitt Decl.) ¶¶ 47–48; Ex. 1004 (Sloan Decl.) ¶¶ 32–33; Opp. 5. Despite this prevailing view in the art—which is reflected in the ’418 Application itself—the ’418 Application “provides no disclosure as to how a skilled artisan could overcome the[] solubility issues and get sufficient amounts of carbidopa (or L-dopa) into solution without using the specific disclosed phosphate prodrugs within Formulas (I) and (II).” Ex. 1004 (Sloan Decl.) ¶ 68; *see also id.* ¶¶ 66–67.

Patent Owner seeks to excuse the lack of disclosure in the application of an LD/CD formulation for use in the claimed methods because “off the

¹⁸ Pfeiffer, *Transdermal drug delivery in Parkinson’s disease*, 3(4) Aging Health 471–82 (2007).

¹⁹ Lundqvist, *Continuous levodopa for advanced Parkinson’s disease*, 3(3) Neuropsychiatric Disease and Treatment 335–48 (2007).

shelf” formulations that could be used in the claimed method were purportedly known, and a “patent ‘preferably omits’” well-known information. *See* Reply 6–7 (citing, *e.g.*, Ex. 2088 (Jenner Decl.) ¶¶ 48–81, 169–174; Ex. 2087 (Fasano Decl.) ¶ 42 (noting that oral administration of levodopa and carbidopa has been the conventional approach for treating Parkinson’s Disease); Ex. 2090 (Remenar), 26–28; Ex. 2093 (Nutt), 483–84; Ex. 1012 (Quinn), 1132; Ex. 2095 (Larson) ¶¶ 48–64); *Falko-Gunter Falkner*, 448 F.3d at 1365–66; *McRO, Inc. v. Bandai Namco Games Am. Inc.*, 959 F.3d 1091, 1102 (Fed. Cir. 2020).

To be sure, the Federal Circuit has stated that to satisfy the written description requirement, “it is unnecessary to spell out every detail of the invention in the specification.” *LizardTech, Inc. v. Earth Res. Mapping, PTY, Inc.*, 424 F.3d 1336, 1345 (Fed. Cir. 2005) (citations omitted). Nevertheless, “enough must be included to convince a person of skill in the art that the inventor possessed the invention.” *Id.* Here, as Petitioner establishes, the ’418 Application “says nothing about ‘off the shelf’ aqueous solutions of LD and CD, much less suggest[s] that they could be used in the claimed methods.” Sur-reply 4; *see also* Ex. 2088 (Jenner Decl.) ¶ 95 (Patent Owner’s declarant conceding that “information about prior art solutions of levodopa and carbidopa is not repeated in the specification”). Again, “the hallmark of written description is disclosure”; “it is the specification itself that must demonstrate possession.” *Ariad Pharms.*, 598 F.3d at 1351, 1352. The critical formulation information Patent Owner cites to support these claims—*e.g.*, an indication that the inventive methods extend to use of off-the-shelf levodopa/carbidopa formulations—is missing here.

In fact, far from indicating that off-the-shelf levodopa/carbidopa formulations can be used in the inventive methods, the '418 Application references existing formulations only to criticize them as insufficient due to solubility issues. *See* Ex. 2060 ('418 Application), 16 (¶ 5) (stating that due to the “low aqueous solubility at the pH range required for infusion,” “[s]table, more highly concentrated, and/or less viscous formulations comprising L-dopa and/or carbidopa (or compounds capable of in vivo bioconversion to L-dopa and/or carbidopa) are desirable”), 23 (¶¶ 49–50) (discussing challenges to achieving suitable levodopa/carbidopa formulations for infusion and introducing the purportedly novel prodrugs as “overcom[ing]” these challenges); *see also* Sur-reply 4 (“[T]he application repeatedly explains the significant challenges presented by LD and CD’s inherently low aqueous solubility.”). The '418 Application nowhere asserts that levodopa/carbidopa formulations of sufficient solubility exist for use in the disclosed methods. *See* Sur-reply 4. A patentee cannot “rely on information that is well-known in the art” when the “four corners of the specification directly contradict information that the patentee alleges is well-known.” *Bos. Sci. Corp. v. Johnson & Johnson*, 647 F.3d 1353, 1366 (Fed. Cir. 2011).

Still further, Patent Owner has not established that the off-the-shelf formulations are even suitable for use in the claimed method. Most of the formulations Patent Owner identifies do not contain both levodopa (or a claimed prodrug) and carbidopa, and none had been administered subcutaneously. *See* Sur-reply 5; Ex. 1091 (Jenner Depo. Tr.), 71:20–72:16; Ex. 1012 (Quinn), 1132 (intravenous levodopa-only solution); Ex. 1025

(Heller²⁰), ¶ 270 (levodopa-only solution); Ex. 2093 (Nutt), 484 (intravenous levodopa-only solution); Ex 2095 (Larson), ¶ 48 (levodopa salt only). The prior art levodopa/carbidopa solutions Patent Owner identifies were used for oral administration, not continuous subcutaneous administration. *See* Sur-reply 5; Ex. 2081 (Kurth²¹), 332, 335; Ex. 2090 (Remenar), 17–18, 40; Ex. 1090 (Fasano Depo. Tr.), 15:15–16:13; Ex. 1091 (Jenner Depo. Tr.), 67:21–68:11.

Patent Owner’s declarant Dr. Jenner testifies that even though the “prior art viewed the ‘low aqueous solubility’ of levodopa and carbidopa as a ‘major challenge,’” the application demonstrates that the inventors purportedly made a “surprising discovery that levodopa and carbidopa together . . . could be dosed via continuous subcutaneous administration to achieve therapeutic levodopa plasma levels greater than about 1000 ng/mL.” Ex. 2088 (Jenner Decl.) ¶ 88. He further testifies that without the work disclosed in the application, “the POSA might still believe, or erroneously think, that levodopa and carbidopa themselves cannot be used by continuous subcutaneous administration.” *Id.* ¶ 89. On this record, we agree with Petitioner that these opinions lack credibility. *See* Tr. 47:17–26.

Dr. Jenner has 40 years of experience in the field. Ex. 2088 (Jenner Decl.) ¶ 129. Yet it was not until his work on this proceeding that he learned of the inventors’ “surprising discovery” that purportedly changed the belief of persons of ordinary skill in the art regarding the subcutaneous infusion of levodopa and carbidopa themselves. *See* Tr. 32:16–22 (Patent Owner’s

²⁰ Heller et al., US 2014/0088192 A1, published March 27, 2014.

²¹ Kurth, *Using Liquid Levodopa in the Treatment of Parkinson’s Disease*, 10(5) *Drugs & Aging* (1997), 332–40.

counsel indicating that Dr. Jenner first changed his view in connection with his work on this proceeding). Indeed, more than six years after the priority date, Dr. Jenner co-authored an article detailing the “major challenges in utilising the subcutaneous route,” including the “lack of solubility of levodopa” and the added complexity of delivery with carbidopa, “with no suggestion that long-known techniques could overcome these challenges.” Sur-reply 8 (quoting Ex. 1086 (Urso²²), 1157).

As recently as 2021, the inventors themselves also continued to identify the poor solubility of levodopa and carbidopa as an obstacle to their subcutaneous delivery. See Ex. 1052 (Voight²³), 1369 (“Unfortunately, the poor solubility of LD/CD leads to a large dose volume, requiring surgery for Duodopa administration.”).²⁴ The inventors thus touted phosphate prodrugs of levodopa and carbidopa—not off-the-shelf levodopa/carbidopa formulations—as a solution to this problem. See *id.* (“We sought a new treatment for PD . . . culminating in the discovery of Foslevodopa/Foscarbidopa.”²⁵); see also Ex. 1047 (Forrest Decl.) ¶ 37. Neither party points to any recognition in the art (e.g., in published articles) of the

²² Urso et al., *Improving the Delivery of Levodopa in Parkinson’s Disease: A Review of Approved and Emerging Therapies*; 34 CNS Drugs 1149–63 (2020).

²³ Voight et al., *Fueling the Pipeline via Innovations in Organic Synthesis*, 12 ACS Med. Chem. Lett. 1365–73 (2021).

²⁴ The Duodopa commercial product is a levodopa/carbidopa gel that is delivered directly to the small intestine via a tube inserted in the stomach wall and connected to an external infusion pump. Pet. 16; Ex. 1001, 2:21–39.

²⁵ Foslevodopa and Foscarbidopa are phosphate prodrugs of levodopa and carbidopa, respectively. See Tr. 9:23–10:6.

supposed breakthrough demonstrated by the '507 patent's disclosure regarding the use of a continuous subcutaneous infusion of levodopa and carbidopa. *See* Sur-reply 1–2. And to date, a continuous subcutaneous infusion of levodopa and carbidopa has not been commercialized.²⁶ *See id.*; Ex. 1003 (LeWitt Decl.) ¶ 48.

On this record, Petitioner convinces us that Patent Owner's arguments regarding the alleged written description support for a formulation comprising levodopa and carbidopa amount to a post-hoc justification designed to secure overbroad claims. Sur-reply 1–2. This is exactly the scenario the written description requirement seeks to guard against. *See, e.g., Ariad*, 598 F.3d at 1353–54 (“[T]he purpose of the written description requirement is to ensure that the scope of the right to exclude, as set forth in the claims, does not overreach the scope of the inventor's contribution to the field of art as described in the patent specification.”) (citations omitted); Pet. 55.

Even beyond finding that the '418 Application fails to disclose the use of carbidopa/levodopa formulations in the claimed method, we further find that the '418 Application does not demonstrate possession of the administration of such a formulation for *continuous* subcutaneous administration, as recited in proposed substitute claim 18. *See* Opp. 7–8.

²⁶ Petitioner asserts that it has “developed a proprietary formulation, known as ND0612, which “combines L-dopa and carbidopa with specific excipients in a way that overcomes the solubility issues and allows sufficient L-dopa and carbidopa to enter into solution for continuous subcutaneous administration.” Pet. 20; Ex. 1003 (LeWitt Decl.) ¶ 49. Patent Owner acknowledges that the ND0612 “is both proprietary and undisclosed to date.” Mot. 19.

Examples 16 and 19 in the '418 Application state that the dosing route was “subcutaneous,” but there are multiple ways to administer a drug subcutaneously, including continuously or via a bolus injection. *See, e.g.*, Ex. 2088 (Jenner Decl.) ¶ 82. Dr. Forrest explains why the administration in these examples was likely a bolus injection. Ex. 1047 (Forrest Decl.) ¶¶ 62, 66, 86–87. Bolus injections can allow for larger volumes than are feasible for continuous subcutaneous administration. *See* Opp. 7–8; Ex. 2087 (Fasano Decl.) ¶¶ 47–48.

Patent Owner does not dispute that Examples 16 and 19 disclose bolus, rather than continuous, subcutaneous injections of carbidopa and levodopa. *See, e.g.*, Reply 5–6; Ex. 2086 (Rosenbaum Decl.) ¶ 51 (conceding that Dr. Forrest’s opinion that Example 16 likely involved a bolus injection “may be a reasonable conclusion”); ¶ 59 (“it is reasonable to assume that the data provided in Table 19-A was generated via a short subcutaneous injection (i.e., bolus)”). Nor does Patent Owner point to any disclosure in the application that expressly discloses *continuous* subcutaneous administration of levodopa and/or carbidopa. Instead, Patent Owner argues that “a subcutaneous bolus injection just uses a shorter period of time,” and “the resulting data” is applicable “to longer periods (i.e., continuous infusions).” Reply 5 (citing, *e.g.*, Ex. 2087 (Fasano Decl.) ¶¶ 47–48; Ex. 2086 (Rosenbaum Decl.) ¶¶ 24, 26, 51, 59, 87; Ex. 2075 (Broadhead²⁷), 326).

In particular, Patent Owner appears to suggest that a person of ordinary skill in the art would have been able to use the pharmacokinetic

²⁷ Broadhead et al., *Parenteral Dosage Forms*, in *Pharmaceutical Preformulation and Formulation* 325–347 (Gibson ed. 2nd ed. 2009).

data in the application generated from the bolus subcutaneous injections of levodopa and carbidopa in order to achieve continuous subcutaneous administration of those compounds. *See* Reply 5–6. According to Patent Owner, in order to practice the claimed method, a person of ordinary skill in the art could have used the pharmacokinetic data to develop a suitable volume of a formulation containing both levodopa and carbidopa, devise a suitable rate of administration of that formulation, and ascertain a suitable number of administration sites. *See id.* (arguing that the “compositions can be subcutaneously administered, e.g., via multiple injection sites of 2–3 mL/hr to meet the calculated administration rate,” and “the volumes are feasible”) (citing, e.g., Ex. 2088 (Jenner Decl.) ¶¶ 23, 83, 84, 86, 170, 185–91; Ex. 2087 (Fasano Decl.) ¶¶ 47–48, 51).

As an initial matter, on this record, Patent Owner does not convince us that a person of ordinary skill in the art would have gleaned all of the suggested information based on the disclosure, given the application’s teachings about the solubility concerns plaguing development of a levodopa/carbidopa formulation for continuous subcutaneous administration. But even if a person of ordinary skill in the art could have or would have done so, the application at best merely renders the claimed method obvious, because none of what Patent Owner describes is found within the application itself. *See* Sur-reply 3–5 (correctly arguing that Patent Owner “improperly bases its arguments entirely on evidence *outside the application*”). And again, “a description that merely renders the invention obvious does not satisfy the [written description] requirement”; “it is the specification itself that must demonstrate possession.” *Ariad Pharms.*, 598 F.3d at 1352. Critical information to support these claims—e.g., an indication that the inventive methods extend to use of off-the-shelf levodopa and carbidopa

formulations, and that those formulations can be used for continuous subcutaneous administration as claimed—is missing here.

For all of the above reasons, we find that Petitioner demonstrates by a preponderance of the evidence that the application does not provide adequate written description support for the claimed formulation limitations [18a–c], particularly as they relate to a formulation comprising carbidopa itself with levodopa (or one of the three claimed levodopa prodrugs), and to such a formulation suitable for continuous subcutaneous administration.

2. *Whether the '418 Application Describes the Claimed Pharmacokinetic Limitations ([18d], [18f])*

Petitioner asserts that because the '418 Application “fails to disclose even a single example of the claimed formulations,” it “fails to disclose the claimed steady state levodopa plasma levels resulting from administration of such a formulation.” Opp. 10.

Patent Owner responds that Examples 16–21 in the '418 Application build on each other, providing *in vivo* pharmacokinetic information from subcutaneous administration, culminating in Example 24, which can be used to predict human steady state levodopa plasma levels and corresponding doses for continuous subcutaneous administration. *See* Reply 2–4 (citing, e.g., Ex. 2086 (Rosenbaum Decl.) ¶¶ 12, 50–67, 69–71, 90–110, 112–22).

Example 24 nominally “purports to predict the steady state plasma exposures of levodopa in humans following administration of a formulation including two prodrugs, carbidopa 4' monophosphate and levodopa 4'-monophosphate.” Opp. 11. Patent Owner, however, argues that a person of ordinary skill in the art “would have understood that the prediction in Example 24 was not limited to only one LD/CD combination, but illustrative of other combinations within the scope of the invention, because all the

claimed compounds were sufficiently characterized individually (Examples 16 and 19-A) and in representative combinations (Examples 17 and 19-B²⁸).” Reply 4 (citing, *e.g.*, Ex. 2086 (Rosenbaum Decl.) ¶¶ 85–92). In other words, as Dr. Rosenbaum opines, a person of ordinary skill in the art “would readily appreciate that the steady state plasma estimation taught in Example 24 could be applied to other [undisclosed] combinations of the compounds identified and tested in the ’507 patent,” and from that information, a person of ordinary skill in the art could “determine how much of a particular combination of the compounds, for which pharmacokinetic data or information was provided, should be administered by continuous subcutaneous infusion to achieve the blood plasma concentrations in the proposed substitute claims of the ’507 patent.” Ex. 2086 (Rosenbaum Decl.) ¶ 12.

After consideration of all of the arguments and cited evidence of record, we agree with Petitioner that the ’418 Application fails to provide adequate written description support for the pharmacokinetic limitations of proposed substitute claim 18 (i.e., limitations [18d], [18f]). Petitioner establishes that “[t]he application does not include, or even suggest performing, PK simulations for the claimed formulations. Indeed, the ’418 Application does not suggest simulating *any formulations* other than the 4’ monophosphate prodrugs in Example 24.”²⁹ Sur-Reply 4. The simulations

²⁸ Examples 17 and 19-B describe pharmacokinetic studies in rats and minipigs, respectively. Ex. 2086 (Rosenbaum Decl.) ¶¶ 56, 60. In both studies, a combination of a levodopa prodrug was administered together with various doses of a carbidopa prodrug. *See id.*

²⁹ Patent Owner criticizes Petitioner’s declarants for not performing any pharmacokinetic models based on the data in the application. Reply 3

Dr. Rosenbaum discusses “cannot support the PK elements because they are entirely outside of (and nowhere suggested by) the application and, at best, would render the PK elements obvious.” Sur-reply 5; *see also Ariad Pharms.*, 598 F.3d at 1353 (“A patent is not a hunting license. It is not a reward for the search, but compensation for its successful conclusion.”).

For the above reasons, we find that Petitioner demonstrates by a preponderance of the evidence that the ’418 Application does not provide adequate written description support for the claimed pharmacokinetic limitations ([18d], [18f]), particularly as they relate to a formulation comprising carbidopa itself with levodopa (or one of the three claimed levodopa prodrugs). This is an additional, independent reason why we determine that Petitioner has shown by a preponderance of the evidence that the proposed substitute claims are unpatentable.

3. *Conclusion*

For the reasons discussed above, on the full trial record, we find that Petitioner demonstrates by a preponderance of the evidence that the proposed substitute claims are unpatentable for lack of adequate written description support.

G. *Petitioner’s Remaining Arguments*

Petitioner additionally argues that the proposed substitute claims are unpatentable as follows: (1) claims 18–30 lack enablement; (2) claims 18–20

(asserting that “[Petitioner’s] experts admit they did not conduct any similar PK analysis, despite this being the crux of the claims,” and citing Ex. 2063 (LeWitt Depo. Tr.), 235:9–236:7; Ex. 2070 (Forrest Depo. Tr.), 33:20–34:4). Patent Owner, however, does not demonstrate where the application teaches or suggests that a person of ordinary skill in the art should do modeling in order to practice the inventive methods with the claimed compounds.

and 23–30 are anticipated by Yacoby-Zeevi;³⁰ (3) claims 18–20 and 23–30 would have been obvious over Yacoby-Zeevi; and (4) claims 18–30 would have been obviousness over Yacoby-Zeevi and Nyholm.³¹ Opp. 16–25.

As discussed above, we determine that (i) Patent Owner has not carried its burden of showing by a preponderance of the evidence that proposed substitute claims 18–30 introduce no new matter relative to the original disclosure; and (ii) Petitioner has shown by a preponderance of the evidence that proposed substitute claims 18–30 are unpatentable for lack of adequate written description support. Because these determinations are dispositive of Patent Owner’s Motion to Amend as to all of the proposed substitute claims, we need not reach Petitioner’s enablement, anticipation, and obviousness arguments for these claims. *See SAS Inst. Inc. v. Iancu*, 138 S. Ct. 1348, 1359 (2018) (holding that a petitioner “is entitled to a final written decision addressing all of the claims it has challenged”); *Bos. Sci. Scimed, Inc. v. Cook Grp. Inc.*, 809 F. App’x 984, 990 (Fed. Cir. 2020) (nonprecedential) (agreeing that the Board has “discretion to decline to decide additional instituted grounds once the petitioner has prevailed on all its challenged claims”).

³⁰ Yacoby-Zeevi, et al., *Maintenance of Constant Steady State Therapeutic Plasma Concentrations of Levodopa Following Its Continuous Subcutaneous Administration with Carbidopa* (Ex. 1005).

³¹ Nyholm, et al., *Pharmacokinetics of Levodopa, Carbidopa, and 3-O-Methyldopa Following 16-hour Jejunal Infusion of Levodopa-Carbidopa Intestinal Gel in Advanced Parkinson’s Disease Patients*, 15(2) *The AAPS Journal* 316–23, April 2013 (Ex. 1006).

III. PETITIONER’S MOTION TO STRIKE

Petitioner filed a Motion to Strike the Rosenbaum Declaration (Ex. 2086), Fasano Declaration (Ex. 2087), and Jenner Declaration (Ex. 2088), along with “35 new exhibits” submitted together with these Declarations. Paper 49 (“MTS”), 1. As the moving party, Petitioner bears the burden of establishing that it is entitled to the requested relief. 37 C.F.R. § 42.20(c).

1. Exhibits 2066–2085, 2089–2107

As an initial matter, we deny Petitioner’s motion as to exhibits other than the Rosenbaum, Fasano, and Jenner Declarations. Petitioner mentions that its request to strike encompasses “35 new exhibits,” but it fails to specifically identify those exhibits (e.g., by exhibit number). MTS 1. We observe that on June 12, 2023, in addition to filing the Rosenbaum, Fasano, and Jenner Declarations, Patent Owner filed thirty-five other exhibits (i.e., Exhibits 2066–2085, 2089–2107). Accordingly, we will assume that Petitioner seeks to strike Exhibits 2066–2085 and 2089–2107.

Not all of these exhibits, however, appear to have been submitted in support of the Declarations. For example, Exhibit 2070 is a deposition transcript of Dr. Forrest, one of Petitioner’s Reply declarants. *See, e.g.*, Ex. 1047 (Forrest Decl.). Several other of these exhibits also appear to relate to Dr. Forrest. *See, e.g.*, Exhs. 2067 (transcript of Forrest deposition in a different proceeding), 2068 (Forrest declaration from a different proceeding), 2103 (same). Petitioner’s motion does not specifically address Exhibits 2066–2085 and 2089–2107 or explain why they should be stricken, particularly those that appear to be related to Dr. Forrest’s cross-examination.

Accordingly, Petitioner's lack of specificity in articulating why we should strike Exhibits 2066–2085 and 2089–2107 warrants denying Petitioner's motion as to those exhibits.

2. *Rosenbaum, Fasano, and Jenner Declarations*

Turning to the Rosenbaum, Fasano, and Jenner Declarations, Petitioner first argues that they should be stricken because they purport to support arguments Patent Owner made in its Motion to Amend, including arguments on which Patent Owner bears the burden of persuasion, and arguments addressed in the Motion to Amend but supported only by attorney argument. *See, e.g.*, MTS 2–3, 5. We agree with Petitioner that if Patent Owner had sought to rely on the Declarations to support its arguments on issues for which Patent Owner bears the burden of proof in its Motion to Amend (i.e., the statutory and regulatory requirements associated with the Motion to Amend), the Declarations would have been filed too late for this purpose. *See* TPG 74–75 (indicating that it is “improper for a reply to present new evidence (including new expert testimony) that could have been presented in a prior filing”).

However, we do not understand Patent Owner to be relying on the Declarations in an effort to meet the statutory and regulatory requirements associated with its Motion to Amend. Indeed, Patent Owner's Reply nowhere refers to the statutory and regulatory requirements. Instead, the Reply focuses only on Petitioner's alleged grounds of unpatentability raised in Petitioner's Opposition to the Motion to Amend. *See, e.g.*, Reply (i) (Table of Contents).

Our rules and guidelines permit Patent Owner to submit new evidence, including declarations, with its reply to Petitioner's opposition. *See* 37 C.F.R. § 42.23(b); Notice Regarding a New Pilot Program

Concerning Motion to Amend Practice and Procedures in Trial Proceedings under the America Invents Act before the Patent Trial and Appeal Board, 84 Fed. Reg. 9,497, 9,500 (Mar. 15, 2019) (“Generally speaking, new evidence (including declarations) may be submitted with every paper in the MTA process, except a sur-reply.”). This, however, is not the end of the inquiry. Petitioner argues that even though a patent owner is permitted to submit declarations with its reply in support of a motion to amend, the Declarations here are nevertheless untimely because they purport to support arguments Patent Owner articulated in its Motion to Amend, but failed to substantiate with evidence. *See, e.g.*, MTS 2–3, 5.

After considering the arguments and cited evidence of record, we observe that Patent Owner’s Declarations contain material that is both timely and untimely. For example, Patent Owner identifies opinions in the Declarations that are proper response, such as arguments and evidence responsive to material in the Forrest Declaration, which Petitioner submitted with its Opposition to Patent Owner’s Motion to Amend. *See* MTS Opp. 4–5.

On the other hand, Petitioner shows that Patent Owner took positions in its Motion to Amend that it only later sought to support with technical declarations. For example, in the Motion to Amend, Patent Owner elected to assert “that levodopa and carbidopa could be ‘interchanged’ in a formulation with their prodrugs, and that a skilled artisan could ‘perform a pharmacokinetic analysis to predict steady state levodopa plasma levels’ using the ‘parent compounds’ based on Example 24.” MTS 2 (citing Mot. 12–13). Patent Owner, however, “chose to forgo any expert support” on these issues at the time it filed its Motion to Amend, and instead waited until the Reply to submit technical declarations in support of these arguments. *Id.*

We agree with Petitioner that although Patent Owner “was not required to anticipate the arguments that [Petitioner] would make in its Opposition” to the Motion to Amend, if Patent Owner “wanted to provide expert testimony to support the positions it **actually took**” in that Motion, “the TPG required [Patent Owner] to do so at that time, instead of waiting until [Petitioner] had limited ability to respond.” *Id.* at 5. Accordingly, we agree with Petitioner that in this respect at least some portions of Patent Owner’s Declarations are untimely, because it is “improper for a reply to present new evidence (including new expert testimony) that could have been presented in a prior filing.” TPG 74.

Nevertheless, an order striking arguments and evidence in their entirety is an “exceptional remedy,” particularly under the circumstances here, where some portions of the Declarations are proper, while other portions are untimely. TPG 80. Moreover, “there is a strong public policy for making all information filed in an administrative proceeding available to the public.” *Liberty Mut. Ins. Co. v. Progressive Cas. Ins. Co.*, CBM2012-00010, Paper 59 at 40 (PTAB Feb. 24, 2014). Under the facts of this case, Patent Owner was permitted to file evidence to respond to Petitioner’s unpatentability claims, but also belatedly supported positions it made in earlier attorney argument. Petitioner’s Motion to Strike, however, challenges all of Patent Owner’s evidence, without acknowledging that some portions constitute proper reply. Although “[t]he Board is not required to attempt to sort proper from improper portions of the reply” (TPG 74, emphasis added), under the circumstances here, we agree with Patent Owner that Petitioner’s lack of specificity as to what it seeks to strike is fatal to Petitioner’s motion. Accordingly, we decline to strike the Declarations in their entirety as untimely.

Separately, Petitioner argues that we should strike Patent Owner’s “untimely evidence” because it is improperly incorporated by reference into Patent Owner’s Reply. MTS 6–7; *see also* 37 C.F.R. § 42.6(a)(3) (prohibiting incorporating arguments by reference from one document to another). As an initial matter, we first address Petitioner’s argument to the extent it extends to all exhibits filed on June 12, 2023, in addition to the Rosenbaum, Fasano, and Jenner Declarations. We deny such a request out of hand, because Petitioner addresses only the Rosenbaum, Fasano, and Jenner Declarations in detail, and fails to address how any of the other exhibits Patent Owner filed on June 12, 2023, is allegedly improperly incorporated by reference.

As to the Rosenbaum, Fasano, and Jenner Declarations, Petitioner points us to five particular examples of alleged improper incorporation by reference. First, Petitioner points to page 2 of the Reply, where Patent Owner presents a single citation to 85 paragraphs of Dr. Jenner’s Declaration. MTS 6. We agree with Petitioner that this particular citation is overbroad, but instead of striking the entire Jenner Declaration, we simply give this particular citation no weight when evaluating the particular statement Patent Owner makes on page 2 of its Reply in connection with that citation.

Second, Petitioner points to page 9 of the Reply, which cites two portions of the Jenner declaration, one spanning 22 paragraphs and one spanning 16 paragraphs. MTS 6. Third, Petitioner points to page 15 of the Reply, which cites 27 paragraphs of the Fasano Declaration. *See id.* These sections of the Reply respond to portions of Petitioner’s unpatentability arguments that we did not reach. Accordingly, we dismiss Petitioner’s complaints as to these citations as moot.

Fourth, Petitioner asserts that “in multiple places, the Reply cites 19 paragraphs of Dr. Rosenbaum’s declaration.” MTS 7. Petitioner, however, does not indicate where in the Reply these citations occur. Fifth and finally, Petitioner asserts that “there are 19 instances where the Reply cites to 10 paragraphs or more of the one of the three declarations.” *Id.* at 7 (citing Reply at 2, 3, 5, 6, 7, 9, 12, 13, 14, 15). Petitioner did not identify these citations with sufficient particularity to permit us a fair evaluation of the propriety of Patent Owner’s citations.

Petitioner also argues that Patent Owner’s Reply “fails to at all discuss over 130 substantive paragraphs from the three expert declarations.”

MTS 7. Our rules, however, do not require that every statement made in a declaration be cited in a brief. *See, e.g., Sumitomo Elec. Indus., Ltd. v. United Techs. Corp.*, IPR2017-00966, Paper 7, at 11 (PTAB Sept. 13, 2017).

For the above reasons, on this record, Petitioner does not persuade us to strike the Declarations either as improperly incorporated by reference, or for lack of citation in Patent Owner’s Reply.

3. Conclusion

For the above reasons, we deny Petitioner’s motion to strike.

IV. PATENT OWNER’S MOTION TO EXCLUDE

Patent Owner moves to exclude Exhibits 1005, 1007, 1040–1044, 1046, and 1048–1050. Paper 66 (“MTE”). Petitioner opposes the motion. *See* Paper 71. As the moving party, Patent Owner bears the burden of establishing that it is entitled to the requested relief. 37 C.F.R. § 42.20(c).

All of the exhibits that are the subject of Patent Owner’s motion to exclude relate to the prior-art status and/or authenticity of Yacoby-Zeevi and ancillary related documents underlying Petitioner’s prior art-based challenges to the proposed substitute claims. *See generally* MTE. Because

we deny Patent Owner’s request to enter proposed substitute claims 18–30 based on lack of written description support and do not reach Petitioner’s prior-art based challenges to the patentability of these claims, we do not rely on Exhibits 1005, 1007, 1040–1044, 1046, or 1048–1050 in this Decision. Accordingly we dismiss Patent Owner’s Motion to Exclude as moot.

V. CONCLUSION

We grant Patent Owner’s Non-Contingent Motion to Amend as to Patent Owner’s request to cancel claims 1–17 of the ’507 patent. As to Patent Owner’s request to add proposed substitute claims 18–30, based on the arguments and cited evidence of record, we determine that Patent Owner has not carried its burden of demonstrating that the proposed substitute claims have adequate written description support in the original disclosure, and that Petitioner has demonstrated by a preponderance of the evidence that the proposed substitute claims are not patentable because they lack adequate written description support. Accordingly, we deny Patent Owner’s Motion to Amend to the extent it requests to add proposed substitute claims 18–30.

Additionally, we deny Petitioner’s Motion to Strike, and dismiss as moot Patent Owner’s Motion to Exclude.

In summary:

Claims	35 U.S.C. §	Reference(s)/Basis	Claims Shown Unpatentable	Claims Not shown Unpatentable
18–30	112(a)	Written Description	18–30	
18–30	112(a)	Enablement ³²		
18–20,	102	Yacoby-Zeevi ³³		

³² As explained above (*see supra* Section II.G), we do not reach this ground.

³³ As explained above (*see supra* Section II.G), we do not reach this ground.

23–30				
18–20, 23–30	103	Yacoby-Zeevi ³⁴		
18–30	103	Yacoby-Zeevi, Nyholm ³⁵		
Overall Outcome			18–30	

Motion to Amend Outcome	Claim(s)
Original Claims Cancelled by Amendment	1–17
Substitute Claims Proposed in the Amendment	18–30
Substitute Claims: Motion to Amend Granted	
Substitute Claims: Motion to Amend Denied	18–30
Substitute Claims: Not Reached	

VI. ORDER

In consideration of the foregoing, it is hereby:

ORDERED that Patent Owner’s Non-Contingent Motion to Amend is *granted* as to the request to cancel original claims 1–17 of U.S. Patent No. 11,091,507 B2;

FURTHER ORDERED that Patent Owner’s Motion to Amend is *denied* as to the request to add proposed substitute claims 18–30 to U.S. Patent No. 11,091,507 B2;

FURTHER ORDERED that Petitioner’s Motion to Strike is *denied*;

FURTHER ORDERED that Patent Owner’s Motion to Exclude is *dismissed* as moot; and

³⁴ As explained above (*see supra* Section II.G), we do not reach this ground.

³⁵ As explained above (*see supra* Section II.G), we do not reach this ground.

FURTHER ORDERED that, because this is a Final Written Decision, any party seeking judicial review must comply with the notice and service requirements of 37 C.F.R. § 90.2.

PGR2022-00040
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FOR PETITIONER:

William Chad Shear
Geoffrey D. Biegler
Bonnie Fletcher Price
Mark Weinstein
COOLEY LLP
cshear@cooley.com
gbiegler@cooley.com
bfletcherprice@cooley.com
mweinstein@cooley.com

FOR PATENT OWNER:

Anthony Insogna
Matthew Johnson
Sarah Geers
Kunyong Yang
Laura Kanouse
Evan Jones
JONES DAY
aminsogna@jonesday.com
mwjohnson@jonesday.com
sgeers@jonesday.com
kyang@jonesday.com
lkanouse@jonesday.com
evanjones@jonesday.com